
SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INDUSTRY PROBLEM TASK

Artificial Intelligence – Agricultural Yield Prediction and Reinforcement Learning

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Course	Artificial Intelligence
Course Code	CSA17
Assessment	Assessment Tool 1 – Industry Problem Task

Industry Scenario: An agri-tech company has farm records containing soil moisture, rainfall, fertilizer usage, temperature, crop type and past yield. The objective is to predict crop yield and recommend suitable farming actions such as Irrigate, Fertilize, Pest Control or No Action.

Important modelling assumption: The question does not provide an actual farm dataset. Therefore, the model metrics, tree split values and Q-learning values in this report are presented as a **representative worked example** suitable for demonstrating the required AI/ML methodology. They should be replaced with actual computed values if the company supplies the dataset.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning learns a general predictive rule from previously observed labelled examples. For this agricultural problem, historical farm records are used as training examples and the model learns the relationship between environmental/farm-management features and crop yield. A supervised learning approach is appropriate because past yield is known for historical records.

Target variable: Crop Yield (for example, tonnes/hectare). A regression model can predict the continuous yield value. If the company wants a warning system, the yield can also be converted into classes such as Low, Normal and High.

Key features:

- **Soil moisture:** directly affects water availability and plant growth; very low moisture can indicate drought stress.
- **Rainfall:** represents natural water availability and helps explain irrigation requirements.
- **Fertilizer usage:** provides information about nutrient availability; insufficient or excessive use can reduce yield.
- **Temperature:** affects crop growth rate, flowering and stress conditions.
- **Crop type:** different crops have different water, nutrient and temperature requirements.
- **Past yield:** captures historical field productivity and can be a strong baseline predictor.

A Decision Tree is especially suitable for the initial model because it handles nonlinear relationships and produces human-readable rules that can be communicated to farmers.

1(b) Handling Class Imbalance

If Low-Yield/Failure records are much fewer than Normal-Yield records, a classifier may become biased toward the majority class. In a farm advisory system this is dangerous because missing a low-yield warning can lead to crop loss.

Recommended strategy: class weights, optionally combined with SMOTE on the training set. Class weights are a strong first choice because they increase the penalty for misclassifying minority low-yield cases without creating synthetic agricultural records. A suitable weighted loss can give Low Yield a larger weight than Normal Yield.

If the minority class is sufficiently represented, SMOTE can be applied only to the training data after the train/test split. SMOTE should not be applied before splitting because synthetic examples derived from test information can cause data leakage. Random under-sampling is simpler but may discard useful normal-yield examples, so it is less attractive when the agricultural dataset is not very large.

1(c) 70:30 Split and Preprocessing

A **70:30 train-test split** provides enough observations for the model to learn while retaining 30% of the records for an independent evaluation. For an agricultural forecasting use-case, the test set represents unseen field conditions and provides a realistic estimate of generalisation. Where records are time ordered by growing season, a chronological split is preferable to a random split to avoid future information leaking into the past.

Step	Method	Impact
Missing values	Median for numerical variables; mode/explicit values for categorical variables	Reduces loss of information; makes the model robust to incomplete farm logs.
Numerical scaling	Standardisation: $z = (x - \text{mean}) / \text{standard deviation}$	Important for Logistic Regression; not essential for Decision Trees.
Categorical encoding	One-hot encoding for crop type	Converts crop names into numerical indicators without imposing an artificial order.
Outliers	Inspect extreme rainfall, moisture and fertilizer usage; remove only if justified	Prevents skewed data from dominating the model.
Class imbalance	Class weights / SMOTE on training data only	Improves detection of minority low-yield cases.

TASK 2: DECISION TREE FOR CROP YIELD PREDICTION

2(a) Decision Tree Model and First Three Levels

For a classification version of the problem, the target is defined as **Yield Class = Low / Normal / High**. The tree selects the feature that produces the largest reduction in impurity. Using Gini Index, the impurity of a node is $G = 1 - \sum p_i^2$. The best root split is the feature with the largest weighted reduction in Gini impurity.

Representative worked split: In the example dataset, soil moisture gives the largest impurity reduction, followed by rainfall and past yield. Therefore, soil moisture is selected as the root node.

Candidate root feature	Illustrative weighted Gini after split	Illustrative Gini reduction
Soil moisture	0.36	0.18
Rainfall	0.39	0.15
Past yield	0.41	0.13
Temperature	0.44	0.10
Fertilizer usage	0.46	0.08
Crop type	0.47	0.07

First three levels of the representative tree:

Level	Decision rule	Prediction / next split
0 – Root	Soil Moisture ≤ 32%	High-risk branch → Rainfall
0 – Root	Soil Moisture > 32%	Lower-risk branch → Past Yield
1	Rainfall ≤ 420 mm	Low-yield risk
1	Rainfall > 420 mm	Temperature split
1	Past Yield ≤ 2.8 t/ha	Normal/low-risk branch
1	Past Yield > 2.8 t/ha	High-yield branch
2	Temperature ≤ 35°C	Normal yield
2	Temperature > 35°C	Heat-stress risk
2	Past Yield ≤ 3.5 t/ha	Normal yield
2	Past Yield > 3.5 t/ha	High yield

The thresholds above are illustrative. With real data, the tree algorithm automatically chooses thresholds that maximise impurity reduction (classification) or variance reduction (regression).

2(b) Model Evaluation and Missed Low-Yield Warnings

For the representative test set, the following results can be reported for the illustrative Decision Tree classifier:

Metric	Illustrative result	Interpretation
Accuracy	86.7%	Overall proportion of correctly classified cases.
Precision	84.2%	How many predicted positive/high-risk cases were correct.
Recall	81.0%	How many actual low-yield warning cases were detected.
F1-Score	82.6%	Balance between precision and recall.
AUC-ROC	0.89	Good separation between risk classes.

A missed low-yield warning is a false negative: the field is actually at risk but the model predicts it as normal. Agronomically, false negatives are costly because irrigation, nutrient correction or pest intervention may be delayed. To reduce them, the deployment system should use class weights, tune the decision threshold toward higher recall, include time-series sensor readings, and provide an early-warning band rather than relying on a single binary decision.

The advisory should also be validated against agronomist rules. For example, persistently low soil moisture combined with high temperature can justify an irrigation warning even when the machine-learning confidence is moderate.

2(c) Post-Pruning: Cost-Complexity Pruning

A fully grown tree can overfit individual farms or unusual weather records. Cost-complexity pruning removes branches that provide little predictive benefit. The objective can be represented as $R_{\alpha}(T) = R(T) + \alpha|T|$, where $R(T)$ is the tree error, $|T|$ is the number of terminal nodes, and α controls the penalty for complexity.

Model	Tree complexity	Illustrative accuracy	Comment
Pre-pruned	max depth = 5; min samples leaf = 5	85.9%	Controls growth before training finishes.
Post-pruned	Cost-complexity α selected by validation	87.4%	Better generalisation and simpler final tree.

Deployment choice: the post-pruned tree is preferable if validation shows similar or higher recall and accuracy. A smaller tree is easier to explain in a farm advisory app, faster to evaluate, and less likely to encode noise from one particular season or field.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Statistical Model and Comparison

A Logistic Regression model can be used for a binary risk formulation such as **Low Yield Risk = 1** and **Normal/High Yield = 0**. The model estimates $P(Y=1|X) = 1/(1+e^{-(\beta_0+\beta_1x_1+\dots+\beta_nx_n)})$. Numerical variables are standardised and crop type is one-hot encoded.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Decision Tree (illustrative)	86.7%	84.2%	81.0%	82.6%	0.89
Logistic Regression (illustrative)	84.9%	80.5%	83.8%	82.1%	0.90

The illustrative comparison shows that the Decision Tree has slightly higher accuracy, while Logistic Regression has slightly higher recall and AUC. A statistical model is preferable when relationships are approximately linear in the transformed feature space, when calibrated probabilities are important, when the dataset is large and stable, or when coefficients must provide a direct explanation of feature effects.

For a farm advisory application, Logistic Regression can therefore act as a useful benchmark and probability-based risk model, while the Decision Tree can provide intuitive if-then rules.

3(b) Feature Importance Analysis

Rank	Decision Tree – top predictor	Why it matters	Statistical model – top predictor	Why it matters
1	Soil moisture	Strongly separates water-stress conditions.	Soil moisture	Large coefficient magnitude indicates strong risk association.
2	Rainfall	Changes available water and irrigation needs.	Temperature	Heat stress changes low-yield probability.
3	Past yield	Captures field productivity history.	Past yield	Represents persistent field-level productivity.

The two models agree on soil moisture and past yield but rank temperature and rainfall differently. This is reasonable. Decision Trees select variables according to split-based impurity reduction and can model nonlinear thresholds, while Logistic Regression estimates a global linear contribution after encoding and scaling. Feature importance should therefore be interpreted jointly with agronomic knowledge rather than treated as proof of causality.

Practical conclusion: the strongest common predictors are soil moisture and past yield. Their agreement increases confidence that water availability and historical field productivity are important risk signals, while rainfall and temperature provide complementary environmental information.

TASK 4: REINFORCEMENT LEARNING FOR ACTION RECOMMENDATION

4(a) Farming Action Recommendation as an MDP

The recommendation problem can be represented as a Markov Decision Process (MDP) defined by states, actions, rewards and transition probabilities. The objective is to learn a policy that maximises long-term crop yield while controlling resource use and environmental cost.

MDP component	Definition for farming	Justification
States	S1 Seedling, S2 Vegetative, S3 Flowering, S4 Fruiting, S5 Harvest	Each state represents a crop's growth stage, which changes during the season.
Actions	A1 Irrigate, A2 Fertilize, A3 Pest Control, A4 No Action.	These represent the available management decisions.
Reward	Reward = yield improvement score – resource/environmental cost	Encourages higher yield without rewarding wasteful irrigation or excessive fertilization/pesticide use.
Transitions	Action and field condition determine the probability of moving to a next state.	A healthy crop under stress moves to a stressed state because weather and crop response are probabilistic.

A sample reward scale is: +10 for substantial yield improvement, +5 for maintaining a healthy condition, 0 for no measurable change, –5 for avoidable stress, and –10 for severe crop damage. Additional penalties can be assigned for excessive water, fertilizer or pesticide use.

4(b) Q-Learning for Five Growing-Season Episodes

Q-Learning updates the value of a state-action pair using: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. For the worked example, use learning rate $\alpha = 0.5$ and discount factor $\gamma = 0.9$, starting with Q-values of zero.

Episode	State	Action	Reward	Next state	Updated Q(s,a)
1	Vegetative + Dry	Irrigate	+10	Vegetative + Normal	5.00
1	Flowering + Dry	Irrigate	+8	Flowering + Normal	4.00
2	Vegetative + Dry	Irrigate	+6	Vegetative + Normal	5.50
2	Flowering + Normal	No Action	+5	Fruiting + Normal	2.50
3	Vegetative + Dry	Irrigate	+8	Vegetative + Normal	6.75
3	Fruiting + Pest Risk	Pest Control	+9	Fruiting + Safe	4.50
4	Flowering + Normal	No Action	+6	Fruiting + Normal	4.25
5	Fruiting + Pest Risk	Pest Control	+8	Fruiting + Safe	6.25

Example update – Iteration 1: If $Q(\text{Dry Vegetative, Irrigate})=0$, reward=10 and the best next-state value is 0, then $Q = 0 + 0.5(10 + 0.9 \times 0 - 0) = 5.0$.

Example update – Iteration 2: If the same state-action has current $Q=5.0$, reward=6, and the best next-state value is 1.0, then $Q = 5 + 0.5(6 + 0.9 \times 1 - 5) = 5.45$ (approximately).

Example update – Iteration 2 for a second pair: If $Q(\text{Flowering, No Action})=2.5$, reward=6 and $\max Q(\text{next})=3.0$, then $Q = 2.5 + 0.5(6 + 0.9 \times 3 - 2.5) = 4.6$.

The agent is trained for at least five complete growing-season episodes. In a real system, many more episodes would be required. A practical convergence criterion is that the maximum absolute change in the Q-table is below 0.01 for 10 consecutive episodes, while average episode reward remains stable.

4(c) Final Learned Policy

State	Best action	Policy rationale
Seedling + Dry	Irrigate	Protect early growth from water stress.
Vegetative + Dry	Irrigate	Water availability strongly affects biomass development.
Vegetative + Nutrient Low	Fertilize	Correct nutrient deficiency at an effective growth stage.
Flowering + Dry	Irrigate	Avoid stress during a yield-sensitive stage.
Flowering + Pest Risk	Pest Control	Prevent damage to flowers and developing yield.

State	Best action	Policy rationale
Fruiting + Normal	No Action	Avoid unnecessary resource use when crop condition is healthy.
Pre-harvest + Normal	No Action	Minimise unnecessary intervention near harvest.

Ethical and Deployment Considerations

- **Resource overuse:** the agent must not learn that more irrigation or fertilizer is always better. Resource penalties should be included in the reward function.
- **Environmental impact:** excessive fertilizer can contribute to nutrient runoff, while unnecessary pesticide use can harm biodiversity. Safe-use constraints should override the learned policy.
- **Farmer trust and explainability:** recommendations should show the main reason, such as “soil moisture is below threshold,” rather than presenting only an unexplained action.
- **Human oversight:** high-impact decisions should be confirmed by a farmer or agronomist, especially when confidence is low or sensor data are abnormal.
- **Data quality and fairness:** models trained on one crop, soil type or region may not generalise to another. The system should monitor performance across farms and seasons.
- **Safety constraints:** pesticide and fertilizer recommendations should respect agronomic limits, legal requirements, crop labels and environmental conditions.

OVERALL CONCLUSION

The proposed AI solution combines supervised inductive learning, Decision Trees, statistical learning and Reinforcement Learning. The Decision Tree provides interpretable yield-risk rules, Logistic Regression supplies a useful probability-based benchmark, and Q-Learning converts field conditions into sequential farming recommendations. For deployment, the system should prioritise recall for low-yield risk, use post-pruning for interpretability, validate predictions on future growing seasons, and keep a human agronomist or farmer in the decision loop.

Final recommendation: deploy a post-pruned Decision Tree or an ensemble model for yield-risk prediction, supported by a calibrated statistical risk score, while using constrained Reinforcement Learning only for recommendation assistance rather than fully autonomous farm control.